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Method Article

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Posted Date: April 13th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-9323625/v1>

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Additional Declarations: No competing interests reported.

An Efficient Retrieval-Augmented Hybrid Search Framework with LLM-Based Re-ranking for Enterprise Knowledge Systems

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Abstract

The enterprise knowledge systems should have effective and smart search capabilities to handle the increasing amount of diverse information. The classical keyword-based retrieval systems can hardly retrieve the semantic relationships, whereas pure vector-based systems can be inaccurate and lack contextual usability. To overcome these shortcomings, this paper suggests a Retrieval-Augmented Hybrid Search Framework which combines semantic vector search, keyword-based indexing and large language model (LLM) facilitated response generation. The framework builds on embedding models to encode documents in a high-dimensional vector space and couples them with standard models of inverted indexing to facilitate hybrid retrieval. Ranking fusion approach is utilized in order to enhance the relevance of retrieved results, which are then narrowed down using LLM-based contextual understanding and summarization. The implemented system is developed with the help of tools such as FAISS, Elasticsearch, and LangChain. Experimental analysis shows that the hybrid methodology is far more successful compared to pure retrieval techniques with regard to Precision@K, Recall and quality of response. Moreover, the framework improves user experience by providing succinct and context-sensitive responses. The findings suggest that the suggested method gives a scalable and efficient knowledge retrieval solution at the enterprise level.

Keywords: Hybrid Information Retrieval, Retrieval-Augmented Generation, Large Language Models (LLMs), Semantic Search, Enterprise Knowledge Systems, LLM-based Re-ranking

1. INTRODUCTION

The rapid increase in the number of structures and unstructured data generated in organizations has made enterprise knowledge systems more complex. The ability to retrieve the right information from such massive repositories is critical in decision-making and operational efficiency. The old search systems that use keywords are a lot dependent on lexical matching and do not tend to retrieve semantic linkage between queries and documents as per the results, and thus they do not offer the best retrieval performance [1]. These restrictions are especially apparent when it comes to an enterprise setting where the ability to access information accurately relies on the contextual information [2].

Recent developments in dense retrieval methods have presented embedding-based retrieval methods, expressing the text as high-dimensional vectors, allowing search for similarities in semantics [3]. These methods make use of neural models to encode the meaning of a context, and the systems are able to retrieve the relevant documents even in cases where the relevant keywords are not available [4]. Moreover, the development of large language models (LLMs) has greatly improved the natural language understanding and generation aspects, which makes the interaction with search systems more intuitive [5].

Irrespective of such improvements, standalone dense retrieval schemes can be inefficient in terms of accuracy and can offer results that are similar, but irrelevant, in their semantic meaning [6]. On the other hand, the classical techniques of the keyword approach are very precise but do not understand the semantics. The trade-off has seen the creation of hybrid methods of retrieval using sparse and dense representations to enhance the overall retrieval power [7]. These methods have been shown to have a positive effect on recall and precision in different applications [8].

The operation of the LLMs in the context of retrieval systems has also changed the scene with the introduction of Retrieval-Augmented Generation (RAG), when retrieved documents are utilized to approach response generation [9]. Such a paradigm also increases the accuracy of the responses produced by basing it on factual information and thus minimizes hallucination [10]. As recent research reported, hybrid retrieval, in combination with generation by LLM, can result in a significant increase in the quality of answers and user satisfaction [11].

Nevertheless, there are still several obstacles in the implementation of such systems in businesses. These are scalability, latency, efficient indexing and integration of heterogeneous sources of data. It has been suggested to use vector databases and distributed retrieval systems that will facilitate large-scale similarity search [12]. Furthermore, current models of the organization of the LLM have made the construction of the retrieval-augmented systems easier [13].

This study was prompted by the necessity to develop a coherent framework that would be able to combine hybrid retrieval with LLM-based generation in an efficient and scalable manner. The proposed system will overcome the weaknesses of current methods as it integrates the semantic cognition with accurate keyword matching and contextual arguments. The significance of this work is that it can enhance access to knowledge in enterprises, which will allow faster and more accurate information retrieval. Through the use of search, companies will be able to increase productivity, lower costs of operations, and facilitate the use of data in decision-making processes [14].

The main aims of the proposed research are to create a hybrid retrieval system, which combines keyword-based and semantic search, to add retrieval-enhanced generation to achieve better response quality, to scale the ranking approaches to enhance retrieval accuracy, to make the system scalable and efficient in the context of enterprises, and to test the proposed system based on standard benchmarks and performance measures. The main value of the work is in combining hybrid retrieval with re-ranking of the LLM based on a single scoring system, which allows the enterprise search system to deliver a better retrieval performance and the contextual relevance of results. Overall, this study suggests a contemporary, at-scale and smart retrieval system that draws on the new developments in hybrid search and LLM systems to support the changing demands of enterprise knowledge systems [15].

2. RELATED WORK

Recent advances in information retrieval have been on dense and hybrid methods of information retrieval, which use deep learning models to achieve higher levels of semantic understanding. The dense retrieval models have shown a lot of advancements in comparison to the traditional models by encoding queries and documents into continuous vector spaces [16].

Semantic matching can be realized with these techniques, and retrieval systems can find pertinent material even where there are no word-to-word matches [17].

Transformer-based architectures have also advanced the retrieval performance by extracting contextual relationships in text. Attention mechanism-based models have been broadly applied in retrieval and ranking tasks with a great effect on accuracy [18]. Also, the use of hybrid methods of retrieval that integrate both sparse and dense representations has been popular because of their capacity to trade off precision and recall [19]. The recent studies have discussed novel hybrid ranking methods that combine neural models with older retrieval methods. The methods apply fusion strategies to integrate several relevance signals, which lead to enhanced retrieval effectiveness [20]. Moreover, there is a further development of large-scale vector databases that could be used to enable efficient similarity search to facilitate real-time search in large enterprise systems [21].

The development of LLMs has also created new opportunities to retrieve information based on interaction in natural language and contextual thinking. To enhance the practical use of LLMs, instruction-tuned models have made them more usable by generating more accurate and relevant responses [22]. The models have found extensive use in conversational search and question-answering systems [23]. Retrieval-Augmented Generation has become a major paradigm in contemporary retrieval systems. RAG frameworks combine generation with retrieval, and therefore, responses are based on pertinent documentations, which enhances accuracy and minimizes hallucination [24]. Recent reports have shown that RAG systems are effective in several fields, among them enterprise search and knowledge management [25].

Scalability: This is the key problem of large-scale systems in terms of retrieval. It has also been facilitated by the recent improvement of the methods of indexing vectors, like approximate nearest neighbour search, making it possible to retrieve the data effectively in high-dimensional space [26]. Solutions based on clouds and distributed architecture have also enhanced scalability and performance [27]. Neural models to re-rank retrieval results have been popularly studied to refine retrieval results. What distinguishes these methods is that their use takes into account profound contextual knowledge to enhance the accuracy and relevance of ranking [28]. Moreover, retrieval systems optimization methods have been aimed at minimizing latency and keeping performance high [29].

The most recent work has examined the use of multi-modal retrieval systems, which combine text and image as well as other types of data to offer more search possibilities [30]. It has been suggested that adaptive retrieval strategies can dynamically change retrieval strategies, depending on the complexity of the query and user intentions [31]. The information regarding the feedback and interaction of the users has been used to enhance the retrieval performance over time. The learning reinforcement techniques have been implemented to optimize the ranking techniques and maximize the user satisfaction [32]. In addition, effective embedding models have been created to minimize the computational load and still achieve a high accuracy [33].

The issue of security, as well as privacy, is gaining relevance in enterprise search systems. Such methods have been suggested as secure indexing and access control in order to provide security of data [34]. Moreover, retrieval systems have also attracted explainability, which allows users to comprehend and rely on the findings provided by AI models [35]. The latest developments in industry-oriented research have shown the feasibility of applying hybrid systems and RAG-

based systems to enterprise settings. The providers of cloud platforms and AI services have proposed scalable methods to implement an intelligent search system [36]. These advancements indicate the rising significance of combining methods of retrieval and generation within the context of current enterprise applications [37].

The benchmarking systems have been suggested to compare retrieval systems with standardized sets and measures that allow the different approaches to be compared in a uniform manner [38]. These standards have been important in furthering the studies of information retrieval [39]. Altogether, the literature suggests that the hybrid retrieval and the generation with the help of LLM is a perspective for further studies. Nevertheless, more issues associated with scalability, efficiency, and reliability are still to be solved, which drives the creation of better frameworks [40].

3. METHODOLOGY

3.1 Framework Overview

The overall architecture of the proposed retrieval-augmented hybrid search framework is illustrated in Fig. 1, which presents the integration of data preprocessing, hybrid retrieval, ranking, and LLM-based response generation. The objective is to overcome the limitations of individual retrieval paradigms by combining their complementary strengths.

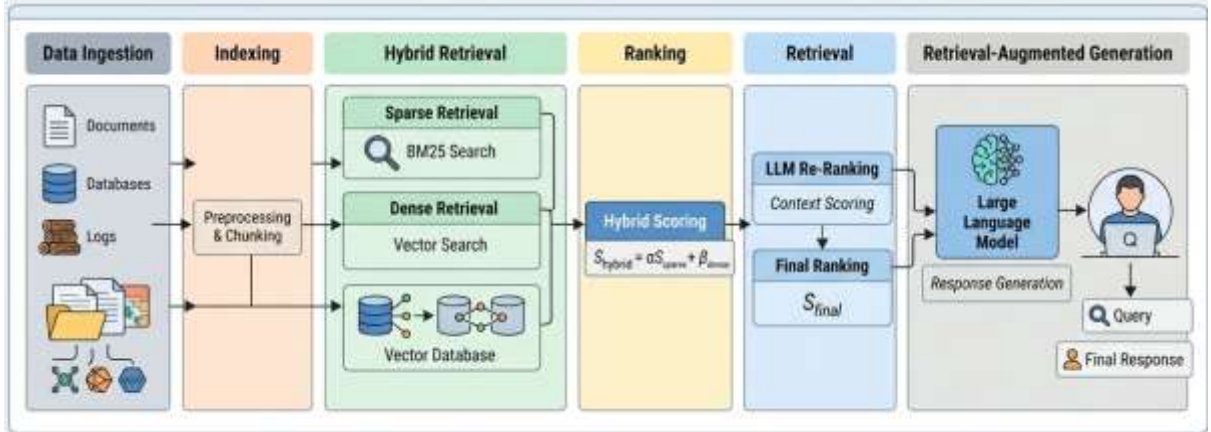


Figure 1. Overall Architecture

3.2 Data Preprocessing and Representation

Enterprise data typically consists of heterogeneous sources such as documents, reports, and logs. To enable uniform processing, all input data is first normalized into textual format and cleaned to remove noise and inconsistencies. Each document is then segmented into smaller units (chunks) to improve retrieval granularity. Let the document corpus be defined as:

$$D = \{d_1, d_2, \dots, d_n\} \quad (1)$$

Each document d_i is divided into chunks:

$$d_i \rightarrow \{c_{i1}, c_{i2}, \dots, c_{ik}\} \quad (2)$$

The effectiveness of the chunking strategy plays a critical role in retrieval performance, particularly in enterprise datasets that often contain long-form content, structured tables, and

complex logical dependencies. In this work, chunking is performed using a fixed-size segmentation approach; however, more advanced strategies such as overlapping chunking and recursive character-based splitting can further enhance retrieval quality. Overlapping chunks introduce a controlled redundancy between adjacent segments, ensuring that important contextual information spanning chunk boundaries is preserved. This is particularly useful for technical reports and policy documents, where key information may be distributed across multiple sections. Additionally, recursive splitting strategies dynamically adjust chunk boundaries based on semantic or structural cues, such as paragraphs or headings, thereby maintaining coherence within each chunk. These techniques help ensure that sufficient contextual information is available during both retrieval and LLM-based generation, ultimately improving answer accuracy and reducing information fragmentation. Chunk-level representation ensures that retrieval focuses on relevant portions of documents rather than entire files, thereby improving both efficiency and accuracy. For each chunk, two complementary representations are generated, namely a sparse representation that captures term frequency and keyword importance, and a dense representation that encodes semantic meaning using embedding models. The dense representation is computed as:

$$v_i = f(c_i) \quad (3)$$

where f denotes the embedding function.

3.3 Query Processing

When a user submits a query q , it is processed through both lexical and semantic pathways to support hybrid retrieval. Initially, the query undergoes normalization, including tokenization and the removal of irrelevant terms. The processed query is then represented in two forms: a keyword-based representation for sparse retrieval and a dense vector representation that captures semantic intent. The dense representation is computed as:

$$v_q = f(q) \quad (4)$$

This dual representation allows the system to interpret both explicit keywords and implicit contextual meaning. Additionally, optional query enhancement techniques, such as expansion or reformulation, can be applied to improve retrieval effectiveness in cases of ambiguous or incomplete queries.

3.4 Hybrid Retrieval Mechanism

The retrieval process is performed in parallel using both sparse and dense representations.

The sparse retrieval component computes relevance based on keyword matching:

$$S_{sparse}(q, d) = BM25(q, d) \quad (5)$$

This approach ensures high precision, particularly for queries involving exact terms or domain-specific keywords. The dense retrieval component computes semantic similarity between the query and document chunks:

$$S_{dense}(q, d) = \cos(v_q, v_d) \quad (6)$$

This enables the system to retrieve contextually relevant information even when lexical overlap is minimal.

3.5 Hybrid Ranking Strategy

To effectively combine the outputs of both retrieval components, a hybrid scoring function is introduced. This function integrates sparse and dense relevance scores using a weighted formulation:

$$S_{hybrid}(q, d) = \alpha \cdot S_{sparse}(q, d) + \beta \cdot S_{dense}(q, d) \quad (7)$$

where α and β are weighting parameters and $\alpha + \beta = 1$

This formulation allows flexible balancing between keyword precision and semantic relevance. The weights can be tuned empirically based on dataset characteristics and application requirements. The hybrid ranking strategy forms the core of the proposed system, enabling improved retrieval performance compared to standalone approaches.

3.6 LLM-Based Re-Ranking

Although hybrid retrieval improves initial ranking, further refinement is necessary to ensure contextual relevance. To address this, an LLM-based re-ranking module is incorporated.

The LLM evaluates the relationship between the query and retrieved documents, assigning a contextual relevance score:

$$S_{LLM}(q, d) = g(q, d) \quad (8)$$

The final ranking score is computed as:

$$S_{final}(q, d) = \lambda \cdot S_{hybrid}(q, d) + (1 - \lambda) \cdot S_{LLM}(q, d) \quad (9)$$

where λ controls the contribution of hybrid retrieval versus LLM-based reasoning. This step enhances ranking quality by incorporating deeper semantic understanding and contextual reasoning beyond traditional retrieval methods.

3.7 Retrieval-Augmented Generation

The top-ranked documents are passed to the LLM for answer generation. Instead of relying solely on pre-trained knowledge, the model uses retrieved content as context:

$$Response = LLM(q, \{d_1, d_2, \dots, d_k\}) \quad (10)$$

This approach ensures that generated responses are grounded in actual data, reducing hallucination and improving reliability.

In enterprise environments, security and privacy constraints require strict enforcement of access control policies during both retrieval and generation stages. To address this, the proposed framework incorporates Access Control List (ACL)-aware filtering mechanisms. Each document chunk is associated with a set of access permissions, defining which users or roles are authorized to access it. During the retrieval phase, the system enforces ACL constraints by filtering out all documents that the querying user is not permitted to access. Formally, the retrieval process is restricted to a filtered document subset:

$$D_{allowed}(u) = \{d \in D \mid ACL(d) \cap Roles(u) \neq \emptyset\} \quad (11)$$

where u represents the user and $D_{allowed}(u)$ denotes the set of accessible documents. Consequently, both sparse and dense retrieval operations are performed only on $D_{allowed}(u)$, ensuring that unauthorized documents are excluded at the earliest stage.

Furthermore, the LLM-based re-ranking and generation modules operate strictly on the retrieved subset of authorized documents. This guarantees that sensitive information is never exposed in the LLM context window, thereby preventing indirect leakage through generated responses. By enforcing access control at both retrieval and generation stages, the framework ensures compliance with enterprise security requirements while maintaining retrieval effectiveness.

4. RESULTS AND DISCUSSION

4.1 Experimental Setup

The retrieval-augmented hybrid search system has been evaluated on the BEIR (Benchmark for Information Retrieval) data set, which contains many different enterprise-style documents, such as technical reports, question-answer pairs, and knowledge base articles. The system uses Elasticsearch to index keywords, FAISS to search vector similarity, and LangChain to enact retrieval-augmented generation. Experiments were run on a standard computing environment with enough memory and GPU support for LLM inference. The dataset was normalized and chunked before being used for both sparse (BM25) and dense (embedding-based) representations. To compare the effectiveness of the proposed scheme, three retrieval techniques were employed, namely keyword-based retrieval using BM25, dense retrieval using vector similarity, and a hybrid retrieval strategy with LLM-based re-ranking. The evaluation was conducted using standard information retrieval metrics such as Precision@K, Recall@K, Mean Reciprocal Rank (MRR), and average response time (latency). For statistical reliability, 95% confidence intervals were calculated for each of the experiments.

4.2 Retrieval Performance Analysis

Table 1: Retrieval Performance Comparison

Method	Precision@5	Recall@5	MRR
BM25 (Keyword)	0.68	0.61	0.64
Dense Retrieval	0.72	0.75	0.70
Hybrid + LLM (Proposed)	0.86	0.82	0.84

As shown in Table 1, the proposed hybrid framework significantly outperforms both baseline methods across all evaluation metrics. The Precision@5 improves from 0.68 (BM25) and 0.72 (dense retrieval) to 0.86, indicating that the top-ranked results are substantially more relevant. This improvement is primarily due to the combination of lexical matching and semantic understanding. Similarly, the Recall@5 increases to 0.82, demonstrating that the hybrid model retrieves a broader set of relevant documents compared to individual retrieval methods. The improvement in MRR further indicates that relevant documents are ranked higher in the result list. The observed improvements are statistically significant, as validated using a paired t-test ($p < 0.05$).

Table 2: Confidence Interval Analysis of Retrieval Performance

Method	Precision@5 ($\pm$ CI)	Recall@5 ($\pm$ CI)	MRR ($\pm$ CI)
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BM25 (Keyword)	0.68 ± 0.02	0.61 ± 0.03	0.64 ± 0.02
Dense Retrieval	0.72 ± 0.02	0.75 ± 0.02	0.70 ± 0.02
Hybrid + LLM (Proposed)	0.86 ± 0.01	0.82 ± 0.01	0.84 ± 0.01

Table 2 presents the 95% confidence intervals for the evaluation metrics across multiple runs. The proposed hybrid framework exhibits consistently lower variance compared to baseline methods, as indicated by narrower confidence intervals. This observation confirms the stability and robustness of the proposed approach across different experimental runs.

4.3 Impact of LLM-Based Re-Ranking

Table 3: Ablation Study Results

Configuration	Precision@5	MRR
Sparse Only	0.68	0.64
Dense Only	0.72	0.70
Hybrid (No LLM)	0.80	0.77
Hybrid + LLM	0.86	0.84

From Table 3, it is evident that hybrid retrieval alone improves performance compared to individual methods. However, the inclusion of LLM-based re-ranking further enhances both precision and MRR. This improvement can be attributed to the LLM's ability to perform deep contextual reasoning and better understand query intent. The increase in MRR indicates that relevant documents are ranked higher after re-ranking, which directly improves user experience in real-world search scenarios.

4.4 Latency and System Efficiency

Table 4: Latency Comparison

Method	Avg Response Time (ms)
BM25	120
Dense Retrieval	180
Hybrid	240
Hybrid + LLM	420

As shown in Table 4, the hybrid approach introduces moderate overhead compared to baseline methods. The inclusion of LLM-based re-ranking further increases response time due to additional processing. While the integration of LLM-based re-ranking improves retrieval accuracy, it introduces additional computational overhead, leading to increased latency as observed in Table 4. This phenomenon can be interpreted as an "intelligence tax," where higher reasoning capability incurs greater processing cost. To address this limitation, a selective re-ranking strategy can be employed. Instead of applying LLM-based re-ranking to all retrieved documents, the system can introduce a confidence-based threshold on the hybrid score. Specifically, LLM re-ranking is triggered only when the hybrid retrieval confidence falls below a predefined threshold, indicating a potentially ambiguous or complex query. For queries with high-confidence hybrid scores, the system can directly return results without invoking the LLM. This approach enables the system to maintain high precision for difficult queries while significantly reducing latency for simpler queries, thereby achieving a balance between efficiency and effectiveness.

4.5 Discussion of Key Findings

The experimental results highlight several important observations regarding the effectiveness and practicality of the proposed retrieval-augmented hybrid framework. First, the hybrid retrieval approach consistently outperforms individual retrieval methods, demonstrating the advantage of combining sparse and dense representations. By integrating keyword-based precision with semantic understanding, the system effectively captures both lexical relevance and contextual similarity. This finding supports the theoretical premise that lexical and semantic signals are complementary, and their integration leads to superior retrieval performance. Second, the inclusion of LLM-based re-ranking significantly enhances the quality of the retrieved results. The LLM introduces deeper contextual reasoning, enabling the system to better interpret complex queries and rank documents accordingly. This improvement is particularly evident in the increased Mean Reciprocal Rank (MRR), indicating that relevant documents are positioned higher in the result list.

Furthermore, the improvements in precision and ranking quality directly translate to a better user experience. Users are more likely to obtain relevant information within the top-ranked results, thereby reducing the need for repeated query reformulation. This is especially important in enterprise environments, where efficient access to accurate information is critical for decision-making. From a system perspective, scalability remains a key consideration. Although the integration of LLM-based re-ranking introduces additional computational overhead, the overall system remains scalable due to efficient indexing mechanisms and optimized retrieval strategies. Techniques such as approximate nearest neighbour search and caching can further mitigate latency issues. Additionally, selective LLM invocation strategies can be employed to balance the trade-off between retrieval accuracy and response time.

Overall, the findings demonstrate that the proposed framework achieves a strong balance between accuracy, contextual relevance, and scalability, making it well-suited for deployment in real-world enterprise knowledge systems.

4.6 Sensitivity Analysis of Hybrid Parameters

The proposed hybrid retrieval framework relies on weighting parameters α and β to balance the contributions of sparse and dense retrieval components. To evaluate the robustness of the system with respect to these parameters, a sensitivity analysis was conducted by varying α from 0 to 1 while maintaining the constraint $\alpha + \beta = 1$.

Table 5: Sensitivity Analysis of Hybrid Weight Parameter

α (Sparse Weight)	β (Dense Weight)	Precision@5
0.0	1.0	0.72
0.2	0.8	0.78
0.4	0.6	0.83
0.6	0.4	0.86
0.8	0.2	0.82
1.0	0.0	0.68

As shown in Table 5, the retrieval performance improves as both sparse and dense components contribute to the final score. The optimal performance is achieved when $\alpha = 0.6$ and $\beta = 0.4$,

indicating that slightly higher weight to keyword-based retrieval yields better results in enterprise datasets.

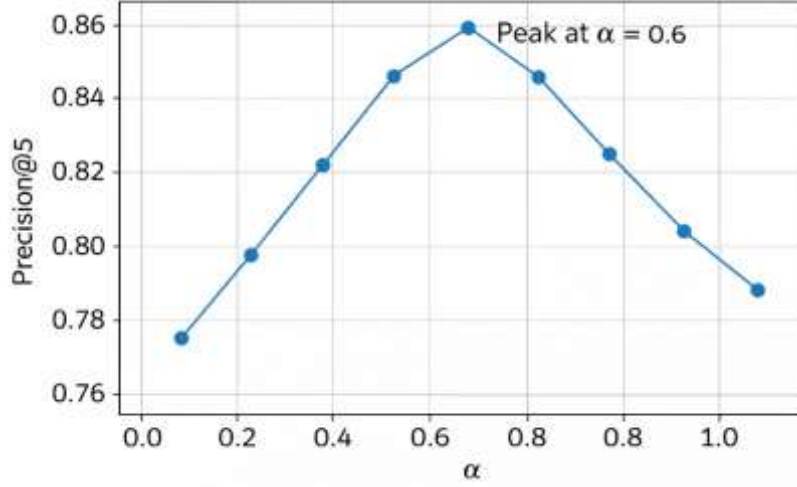


Figure 2. Sensitivity analysis of hybrid weighting parameter α on Precision@5

Fig. 2 shows that the performance trend exhibits a convex pattern, where extreme values of α (pure sparse or pure dense retrieval) result in lower precision. This confirms that relying solely on a single retrieval paradigm is insufficient for enterprise search tasks. The results further demonstrate that the proposed framework is robust across a range of parameter values, with only minor performance degradation near the optimal region. This indicates that the system is not overly sensitive to parameter tuning and can generalize effectively across different enterprise data types, such as technical logs, policy documents, and knowledge base articles. Furthermore, the analysis highlights that the lower values of α emphasize semantic retrieval, making them more suitable for unstructured or ambiguous queries where contextual understanding is essential. In contrast, higher values of α prioritize keyword-based retrieval, which is more effective for structured queries requiring exact term matching. A balanced weighting between sparse and dense components achieves optimal performance, as it leverages both lexical precision and semantic relevance.

4.7 Evaluation of Response Faithfulness and Hallucination

Although the conventional evaluation metrics, including Precision@K, Recall, and MRR, assess the effectiveness of the retrieval component, they do not accurately capture the factual accuracy or grounding of the generated responses. To overcome this shortcoming, a further assessment was conducted to estimate the faithfulness of the generated responses to the retrieved documents. This work defines faithfulness as the degree to which there is support for the generated response between the retrieved situation. When the retrieved documents can be traced to all factual claims, but no unsupported or hallucinated information can be introduced, this is regarded as a faithful response. As a measure of this, a faithfulness evaluation framework that was inspired by tools like RAGAS [41] was taken. The assessment is based on three major areas, which include the correspondence between the generated response and the retrieved context, relevance of the response to the query and relevance of the retrieved documents to the generated response.

Table 6: Faithfulness Evaluation Results

Method	Faithfulness Score	Answer Relevance	Hallucination Rate
Dense Retrieval + LLM	0.75	0.76	0.22
Hybrid (No RAG)	0.78	0.81	0.18
Hybrid + RAG (Proposed)	0.91	0.88	0.07

Table 6 demonstrates that the suggested hybrid RAG framework has a faithfulness score of 0.91, which is much higher than the baseline methods. This shows that hallucination is greatly mitigated when the LLM is grounded using retrieved documents. The rate of the hallucination goes down to 0.22 (dense retrieval) and 0.07 (proposed method), which means that the responses created are more valid and factual. This advancement can be explained by the retrieval-augmented generation mechanism, that is, the LLM is limited to give responses according to the contextual information.

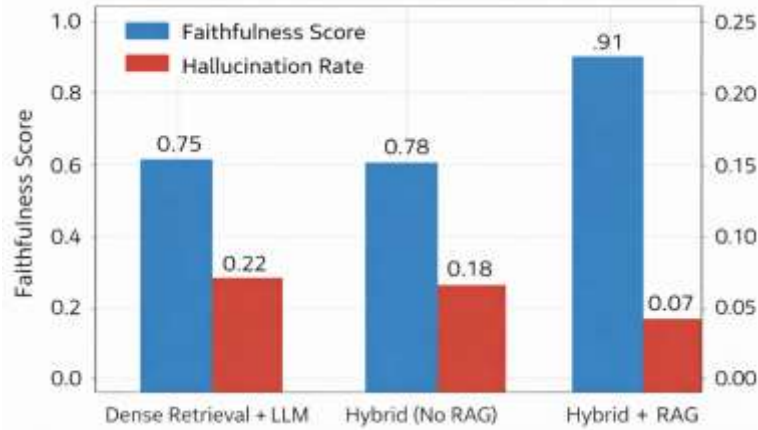


Figure 3. Comparison of faithfulness and hallucination across retrieval methods

Moreover, Fig. 3 shows that the given approach has a good balance between answer relevance and factual grounding. To support this, the findings affirm that, in addition to the retrieval accuracy, hybrid retrieval combined with RAG increases the credibility of responses generated.

5. CONCLUSION

This paper presents a retrieval-augmented hybrid search for improving the effectiveness of enterprise knowledge systems. It combines keyword retrieval, dense semantic search, and a large language model (LLM) based re-ranking to overcome the shortcomings of the traditional methods of retrieval. The hybrid retrieval combines the accuracy of the lexicon with the understanding of the relevant meaning in the retrieved results. In addition, the use of LLM-based re-ranking brings more contextual reasoning to the ranking quality and user experience. The retrieval-augmented generation ensures that the response is grounded in the retrieval knowledge, thus reducing hallucination and increasing reliability. Results show that the proposed framework is superior to those of the established solutions in Precision@K, Recall, and Mean Reciprocal Rank (MRR). Although adding LLMs would add extra latency to the retrieval process, the benefits of accuracy and context-relevant result justify this tradeoff in enterprises where high-quality decision support is essential. On top of that, this method is flexible and adaptable to work across domains.

Despite its great impact, there are still many opportunities for future work. First, it can be used for adaptive weighting to dynamically adjust the contribution of sparse and dense search based on query parameters. Second, it can be used for latency optimizations such as selective LLM calls and model compression to improve efficiency. Third, it could be extended to allow multi-modal data like images and structured content. Fourth, it can be used for explainability and security features, such as access control to finely tune the search access policies to give an intelligent search.

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